

- (c) 'n Waarnemer beweeg met spoed v_r na 'n stilstaande bron met frekwensie f'_0 . Indien die spoed van klank v is, toon aan dat die nuwe frekwensie $f'_1 = f'_0 \left(1 + \frac{|v_r|}{v}\right)$ sal wees.

A receiver moves with velocity v_r towards a stationary source with frequency f'_0 . If the speed of sound is v , show that the new frequency will be $f'_1 = f'_0 \left(1 + \frac{|v_r|}{v}\right)$.

$$\text{new speed} = v' = v + |v_r|$$

$$\therefore f' = \frac{v'}{\lambda} = \left(\frac{v + |v_r|}{v}\right) f'_0$$

$$= \left(1 + \frac{|v_r|}{v}\right) f'_0$$

$$\therefore f' = f'_0 \left(1 + \frac{|v_r|}{v}\right)$$

- (f) Die frekwensie van 'n ambulans se sirene is 1600 Hz. Die ambulans ry teen 95 km/h na die voet van 'n loodregte kran. Die spoed van klank in lug is 330 m/s.
The siren of an ambulance has a frequency of 1600 Hz. The ambulance is driving towards the base of a perpendicular cliff at 95 km/h. The speed of sound in air is 330 m/s.

- (i) Toon aan dat die frekwensie wat die pasiënt, wat van die kran afgeval het, sal hoor 1739 Hz sal wees wanneer die ambulans teen 95 km/h nader kom.
Show that the frequency that the patient, who had fallen from the cliff, will hear if the ambulance is approaching at 95 km/h will be 1739 Hz.

$$f' = f'_0 \left(\frac{1}{1 - \frac{v_r}{v}}\right)$$

$$= 1600 \left(\frac{1}{1 - \frac{95}{330}}\right)$$

$$= 1739 \text{ Hz}$$

- (ii) Bepaal die frekwensie f'_1 van die waarkaatste golf wat die ambulansman sal hoor.
Determine the frequency f'_1 of the reflected sound wave at the ambulance.

$$f'_1 = f'_0 \left(1 + \frac{|v_r|}{v}\right) = 1739 \left(1 + \frac{95}{330}\right)$$

$$= 1878,1 \text{ Hz}$$